
Calibrated Risk Envelopes for LLM Quantization, and the Limits of Predicting Compression Damage

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Abstract

Quantizing a large language model reduces serving cost and costs some accuracy. Determining how much normally requires re-running a full benchmark suite, which is the expense quantization was meant to avoid. We ask whether the accuracy delta can be predicted instead. Using 817 published evaluations spanning 38 base checkpoints, 8 model families and 6 quantization schemes, we report three results. First, a *negative* one: per-model prediction carries almost no signal. Under leave-one-family-out validation the best predictor, a per-scheme mean, beats a global mean by 0.03pp of MAE (95% checkpoint-bootstrap CI [0.01, 0.05]), and both ridge regression and gradient boosting are *worse* than guessing the average, because evaluation noise accounts for at least a third of the variance on 8 of 11 benchmarks. The features we tested are card-level (bit-width, scheme, size, benchmark identity, headroom); we did not test predictors that use checkpoint-derived features such as weight statistics or activation-outlier metrics. Second, a positive one: conditioning only on the quantization scheme yields per-scheme conformal-style intervals with 90.1% two-sided empirical coverage (118/131, 95% CI [83.6, 94.6]) on unseen checkpoints of families that are themselves present in training. Those checkpoints cover 4 of the 6 schemes, and W4A16 supplies 80 of the 131 rows. Each (scheme, size-band) cell returns one of three verdicts: trusted; insufficient evidence, where the interval is still shown but the scheme is demoted and flagged; or refused, where the interval is withheld. Currently 9 of 17 cells are insufficient evidence and 0 are refused. Third, and most consequential for practice: the published corpus is outcome-truncated. Every configuration in it is one that someone chose to release. We rented an A100 and measured deliberately-faulted configurations, observing losses to -39.5pp against a published worst case of -8.86pp —measured, we stress, only on models of 1.5B and below, and on configurations engineered to fail rather than typical ones. This is an existence proof that far worse outcomes exist than the corpus contains; it does not estimate how common they are. We argue the resulting artifact should be described as a calibrated historical baseline rather than a predictor, and we state precisely why its coverage guarantee does not transfer to a practitioner’s own untuned recipe. Code, data and the provenance of every reported figure are provided as supplementary material.

1 Introduction

Post-training quantization is now routine. A practitioner choosing between W4A16, W8A8 and FP8 wants one number: how much accuracy will this cost? In practice the only way to know is to run the evaluation.

37 This paper asks whether that number can be forecast from the public record. Model publishers
38 release quantized checkpoints with benchmark tables comparing the compressed model against its
39 dense parent. Aggregated, these are hundreds of paired measurements of exactly the quantity of
40 interest (850 extracted in our harvest).

41 Our contributions:

- 42 1. **A negative result** (§5). Per-model prediction of the accuracy delta *from card-level features*
43 does not work, and we show why: the target is dominated by evaluation noise. Checkpoint-
44 derived features (weight statistics, activation-outlier metrics) were not tested. We report
45 this as the primary finding rather than burying it.
- 46 2. **A calibrated envelope that does work** (§6), conditioned on scheme alone, with measured
47 coverage on held-out families and a three-state verdict for each (scheme, size-band) cell:
48 *trusted*; *insufficient evidence*, where the interval is still shown but the scheme is demoted
49 and the cell flagged as one the tool cannot judge; or *refused*, where the interval is withheld.
50 Currently 9 of 17 cells are insufficient evidence and 0 are refused.
- 51 3. **An existence proof that the corpus omits severe outcomes** (§7), via GPU experiments on
52 deliberately-faulted configurations—data the published record structurally cannot contain.
53 It shows that losses far beyond anything published are reachable on small models. It does
54 not estimate how large the bias is: our attempt at a correction (in the supplement) did not
55 work well enough to ship.
- 56 4. **An audit** (§9) recording errors we found in our own work, including ones that changed
57 published numbers and claims.

58 2 Related Work

59 **Predicting evaluations.** BenchPress Zeng and Papailiopoulos [2026] establishes the mechanism
60 closest to ours: predict an eval result, estimate whether to trust it, wrap it in a conformal interval, and
61 decide whether to skip the real run. Their score matrix over 84 frontier models and 133 benchmarks
62 is approximately rank-2. We do not claim novelty against them on mechanism. Their estimand is
63 a model’s absolute score; ours is the paired delta between a model and its compressed self, and
64 quantized checkpoints do not appear in their paper.

65 **Conformal prediction on compressed models.** Tong et al. [2026] apply conformal prediction
66 directly to quantized and sparse LLMs, which is a closer domain than BenchPress. The estimand
67 differs: they evaluate predictive uncertainty (coverage and set size) *after* running the compressed
68 model. Ours is constructed over historical accuracy deltas to give an interval *before* running any-
69 thing. Their Figure 2 plots the same quantity we predict, $\text{Acc}_{\text{compressed}} - \text{Acc}_{\text{dense}}$, but measures it.
70 Their finding that compression decouples accuracy from uncertainty bounds what an accuracy-only
71 estimate like ours can claim (§8).

72 **Quantization methods.** The checkpoints we harvest are produced with GPTQ Frantar et al. [2022]
73 and SmoothQuant Xiao et al. [2022] via llm-compressor vLLM Project [2024]. Kumar et al. [2024]
74 fit a functional form that predicts degradation from training and inference in varied precisions, using
75 pretraining runs of models up to 1.7B parameters. We fit no such law: our envelope is empirical, per
76 scheme, over published post-training-quantized checkpoints.

77 **The source of our data.** Kurtić et al. [2024] report over 500,000 evaluations of quantized Llama-
78 3.1 models, finding recovery above 99% of the average OpenLLM v1 score for every scheme tested.
79 As far as we can tell, that work does not address whether the published set of quantized models is a
80 biased sample. Characterising that gap is the contribution of §7.

81 **Selection bias.** Survivorship bias has been identified and measured in a widely used retrieval
82 benchmark Gupta and MacAvaney [2022]. Conformal prediction’s guarantee is known to fail un-
83 der selection: conditional on a selection event, calibration data are no longer exchangeable with a
84 test point Barber et al. [2023], Jin and Ren [2024]. We apply this directly to our own artifact in §8.

85 3 Data

86 We harvest published model cards into rows of (model, scheme, benchmark, $\text{acc}_{\text{before}}$, $\text{acc}_{\text{after}}$), yield-
 87 ing 850 rows, of which 817 remain after dropping 33 whose baseline accuracy is below 20% (near
 88 chance, where a delta is uninformative), across 38 base checkpoints, 8 families and 6 schemes.

89 **An integrity gate.** Cards typically print a *Recovery %* column alongside the before/after pair.
 90 This is redundant information, and we exploit it: for every three-column row we check $\text{recovery} \approx$
 91 $100 \times \text{after/before}$ to a tolerance derived from the card’s own printed precision, and reject rows that
 92 disagree. This caught spurious values that our parser produced under column-order variation be-
 93 tween card templates, and internally inconsistent cards. Rejections are recorded rather than silently
 94 dropped.

95 **A structural limitation, stated at the outset.** All rows come from a single publisher using a
 96 single toolchain. Holding out a model family tests transfer across *models*; nothing in this design
 97 tests transfer across *quantization practitioners*. §7 gives direct evidence that this distinction matters.

98 **The pathway is only sparsely documented.** A live scan on 2026-09-26 across the 439 RedHatAI
 99 quantized-model cards currently published on Hugging Face (of which 102 contribute rows to our
 100 harvested dataset) shows how sparsely the pathway that produced these outcomes is documented:
 101 12 of 439 cards pin a quantization-library version, 33 mention activation reordering, 17 mention
 102 dampening, and 198 of 439 report none of `num_calibration_samples`, `max_sequence_length`,
 103 activation reordering or dampening. On the 102-card subset that our dataset draws rows from, 0
 104 pin a library version and 39 of 102 report none of those four parameters. (This RedHatAI-only
 105 scan is distinct from the 10-publisher, 260-card publisher census reported in §9, which is an earlier
 106 stable snapshot spanning multiple publishers and asks a different question.) The published record we
 107 calibrate against is therefore a record of outcomes with the pathway that produced them substantially
 108 undocumented on the publisher whose cards contribute the largest single share of our corpus, which
 109 is one of the reasons an outcome-truncation argument (§7) cannot be closed from the corpus alone.

110 4 Method

111 **Conformal-style intervals.** We use the residual-quantile construction of split conformal re-
 112 gression Lei et al. [2018]: given residuals $\{s_i\}_{i=1}^n$ and level α , the interval half-width is the
 113 $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score, which under a proper calibration/test split gives exact finite-
 114 sample coverage under exchangeability. Where n is too small for that index to exist, we return
 115 ∞ rather than a misleadingly finite band. The shipped per-scheme interval fits its centre and its
 116 half-width on the *same* rows (an in-sample residual-quantile band), so the split-conformal guaran-
 117 tee holds only for the explicit calibration-set variant used inside the leave-one-family-out audit; the
 118 empirical prospective figure below is what carries the shipped intervals.

119 **Mondrian (group-conditional) calibration.** Calibrating separately within categories is the Mon-
 120 drian construction of Vovk et al. [2005], applied to regression by Boström and Johansson [2020]. A
 121 single marginal interval advertised at 90% covers only 62% of NVFP4 rows under leave-one-family-
 122 out (recomputed live)—the users who most need it. We therefore calibrate per scheme.

123 **A three-state verdict.** For each (scheme, size-band) cell we measure coverage directly, and the
 124 tool returns one of three verdicts. A cell is *trusted* when measured coverage is adequate and ad-
 125 equately supported. It is *refused*—INSUFFICIENT CALIBRATION, no interval—when coverage is
 126 measurably poor: the checkpoint-bootstrap interval on its coverage lies wholly below the 85% line,
 127 on at least 3 checkpoints. It is *insufficient evidence* when the cell rests on fewer than 3 distinct
 128 checkpoints, or when a checkpoint-level cluster bootstrap 90% interval on its coverage straddles the
 129 85% line. In that case the evidence cannot support either of the other two verdicts. The tool still
 130 prints the all-sizes interval for such a cell, but demotes the scheme to the lowest tier and states that
 131 the cell cannot be judged; only a refused cell has its interval withheld.

132 Coverage is scored *one-sided*: how often the true delta stayed at or above the interval’s lower bound.
 133 The risk this tool bounds is accuracy loss, and a model that beats its envelope has exposed no one to

anything, so counting that as a failure would flag cells that are merely outperforming. The interval is symmetric, so a 90% two-sided nominal corresponds to a one-sided nominal near 95%; pooled one-sided coverage across all 817 rows, with each row scored under 7 different calibration families, is 95.1%—essentially at that target, not above it.

Currently 0 of 17 cells are refused. Two cells were flagged under the earlier two-state rule, and neither is a confident refusal under the three-state one. $w4a16|<2B$ had one-sided coverage of 70.1% over 11 rows (two-sided 68.8%), but those rows come from only 2 checkpoints. $w8a16|>10B$ had 81.1% over 31 rows from 5 checkpoints (two-sided 73.7%), and its bootstrap interval, [47, 100]%, straddles the threshold. Both are now *insufficient evidence*. We had scoped the change to those two cells; applying the rule to every cell moved 9 of 17 into that state, including several whose point-estimate coverage looked comfortable. The same measurement and the same rule judge each scheme’s all-sizes coverage: 1 of the 6 schemes is insufficient evidence at that level and 0 is refused. The one-sided scoring and the three-state rule were proposed by an external commenter who identifies as an AI account collaborating with the OpenLanguageModel maintainers; we recomputed each of their figures from the raw rows before adopting either change, and one of their descriptions did not survive that check.

Post-hoc thresholds, honestly labeled. The three-state rule was proposed for two specific cells and then applied to every cell; the particular thresholds—85% for the refusal line, 3 checkpoints for the support floor, a 90% cluster bootstrap for the straddle test—were not derived from a first-principles argument, they were the values under which those two cells came out the way we thought they should. On this corpus one further consequence of that choice is that the *refused* state, as defined, does not fire: every cell with enough checkpoints to reach the second gate has a bootstrap upper bound comfortably above 85%, and the one cell whose upper bound does sit below it ($w4a16|<2B$, upper bound 76.2%) has too few checkpoints to reach that gate and lands in *insufficient evidence* instead. So the current three states are, on this corpus, effectively two: *trusted* and *insufficient evidence*. This is a design finding, not a runtime bug, and we state it here rather than let a reviewer discover it.

A support floor. Coverage measured on many benchmarks from one checkpoint is not independent evidence; it is one quantization run viewed many ways. Cells resting on fewer than 3 distinct checkpoints cannot reach the top tier regardless of measured coverage. 4 cells are affected.

5 The negative result

Under leave-one-family-out validation, row-weighted MAE:

	predictor	MAE (pp)	beats global mean?
	per-scheme mean (shipped)	0.7227	yes
	global mean (baseline)	0.7545	—
	per-(scheme $\times$ benchmark)	0.7607	no
	ridge regression	0.7639	no
	per-benchmark mean	0.7647	no
	gradient boosting	0.7878	no

The per-scheme mean’s advantage is small: 0.03pp, with a checkpoint-bootstrap 95% CI of [0.01, 0.05]pp, and it is positive in 7 of the 8 held-out families. It is a distinguishable but modest gain. Ridge regression ($\alpha = 3.0$, standardised features, no cross-validated tuning) and histogram gradient boosting (max_depth=3, max_iter=300, learning_rate=0.05, min_samples_leaf=15, l2_regularization=1.0, random_state=0, no cross-validated tuning) are both *worse* than predicting the global average. Adding model size, bit-width, method, benchmark identity and headroom each degraded out-of-family performance. All features tested are card-level; we did not compute any checkpoint-level statistic (weight kurtosis, activation outliers, singular-value structure) and feed it in as a predictor. Those directions are open.

Why. Estimating per-benchmark evaluation noise from near-lossless schemes—whose true delta should be approximately zero—gives a noise floor that accounts for anywhere from 15% of the

178 observed variance to all of it across 11 benchmarks. On GSM8K the deltas of supposedly lossless
 179 schemes have a standard deviation of 2.1pp, while the quantity being predicted has mean -0.39pp .
 180 The target is a difference of two noisy quantities that largely cancel: the numerically smallest and
 181 hardest-to-predict summary of quantization damage available. Two cautions on the noise figures.
 182 They assume near-lossless schemes have a true delta near zero, so they are upper bounds on noise.
 183 And 5 of the 11 estimates rest on 10 or fewer rows; one, at 124%, exceeds the total variance, which
 184 we attribute to sampling error rather than treat as a finding.

185 **Implication.** From card-level features, the entire fitted artifact is 18 numbers. A raw per-scheme
 186 historical quantile band, with no model and no conformal machinery, is a serious competitor: it
 187 looked better covered in-sample, though under the strict protocol that advantage disappears (§9).
 188 What conformal contributes is a finite-sample guarantee and a principled rule for small- n schemes,
 189 not accuracy. We therefore describe the result as a *calibrated historical baseline*, not a predictor.

190 6 Results

191 **What this tool is for.** The scenario it answers is: a practitioner is considering a scheme on a
 192 checkpoint from a family already represented in the published record and wants a calibrated range
 193 for the accuracy delta with an honest verdict on whether the record supports that range for that
 194 cell. Within that scope, each cell returns an interval and one of three tier labels; outside it, the tool
 195 declines to speak. It is not a scheme-safety certificate, does not bound a scheme’s worst case, and
 196 does not extrapolate to recipes or families the corpus has not already seen – §7 and §8 state each of
 197 those out-of-scope cases directly.

	scheme	90% interval (pp)	worst observed	% losing >3pp	evals
	fp8	[-1.11, +0.48]	-1.90pp	0.0%	80
	w8a8_int	[-1.71, +1.07]	-3.32pp	1.0%	194
198	fp8_dynamic	[-1.44, +1.34]	-8.72pp	1.0%	200
	w8a16	[-0.98, +0.81]	-3.04pp	1.2%	86
	w4a16	[-2.87, +1.40]	-8.86pp	5.7%	193
	nvfp4	[-4.10, +1.87]	-8.12pp	15.6%	64

199 **Prospective validation.** After freezing the method we harvested new checkpoints not used during
 200 development. Under a strict protocol that excludes any card whose format informed a parser fix
 201 (families Llama-3.1, Llama-4 and Qwen3), two-sided coverage is $118/131 = 90.1\%$, 95% Clopper-
 202 Pearson CI [83.6, 94.6], against a nominal 90% two-sided level. This is in-family transfer: every
 203 strict prospective family is also present in the training corpus. It measures whether the envelope
 204 generalises to unseen *checkpoints* of families it has already seen, not to unseen families. A from-
 205 scratch, standard-library-only reimplementation sharing no code with the pipeline reproduces this
 206 figure with zero disagreements. That checks the computation, not the validity of the design.

207 **What the prospective set does and does not cover.** The 131 strict rows come from 19 quantized
 208 checkpoints, not 131 independent observations: a bootstrap that resamples checkpoints gives a 95%
 209 interval of [83.2, 94.7]%, close to the row-level one. The rows cover only 4 of the 6 schemes; FP8
 210 and NVFP4 have no prospective rows, so the figure says nothing about them.

	scheme	strict rows	checkpoints	covered (%)
	w4a16	80	11	87.5
211	w8a8_int	36	5	91.7
	fp8_dynamic	11	2	100.0
	w8a16	4	1	100.0

212 W4A16, the most common scheme, dominates the pooled figure and sits below the nominal 90%;
 213 the other schemes rest on few checkpoints. Over all 186 harvested prospective rows, before the strict
 214 filter, coverage is 90.3%, so the filter did not manufacture the headline. That said, the pooled 90.1%
 215 is pulled up by 11 FP8-dynamic rows from 2 checkpoints and 4 W8A16 rows from 1 checkpoint,

216 both at 100.0% coverage: W4A16, at 80 of the 131 strict rows (61%), comes in at 87.5%, which is
217 below the nominal 90%. The pooled number should be read as an average across schemes with very
218 different sample sizes, not as a coverage claim uniform across schemes.

219 **What the envelope cannot do.** 0 of 6 intervals exclude zero: for most schemes the true effect is
220 small enough that a calibrated interval built on this record cannot rule out zero damage. And 19 of 25
221 observed losses worse than 3pp fell *below* their interval floor: a symmetric band fit to a left-skewed
222 damage distribution systematically under-covers the tail, and an asymmetric construction that does
223 not degenerate to infinite intervals (§9) is the obvious next step.

224 7 The measured left tail

225 The corpus contains only configurations someone chose to publish. To measure what it structurally
226 cannot show, we rented an A100 and quantized real models under deliberately-faulted recipes.

227 **Result.** Worst measured loss -39.5pp , against -8.86pp as the worst loss anywhere in the pub-
228 lished corpus. 16 of 36 faulted rows lost 3pp or more; separately, 2 of 9 rows from our correctly-
229 configured control arm did too. The dominant single effect was wrong scale granularity—per-tensor
230 instead of per-group.

231 **Scope, stated plainly.** This tail was measured only on models of 1.5B and below. It establishes
232 that catastrophic damage is reachable; it is not a measured bound for larger models.

233 **A confound we found and did not resolve.** Re-running MMLU under the published protocol
234 (5-shot, letter-scored) recovers accuracies in the range published on the cards: our Qwen2.5-0.5B-
235 Instruct scores 47.20 against the Instruct card’s 46.83 (the base-model card gives 47.57), and our
236 Qwen2.5-1.5B-Instruct scores 59.57. The harness question closed. Our control run then compared
237 Instruct-model checkpoints against Red Hat’s *base-model* W4A16 publications, because no Instruct
238 W4A16 checkpoints are published: our correctly-configured Instruct control is $2.6\times$ more damag-
239 ing than the base-model published W4A16 (mean gap -3.78pp , $z = 2.9$ and 5.3). This is not like-
240 for-like—Instruct models could plausibly be more fragile under 4-bit quantization than their base
241 counterparts—so the ratio is indicative rather than a clean estimate. A matched Instruct-vs-Instruct
242 control is deferred future work. The pattern we observed is consistent with approximately $171\times$ less
243 calibration data than production defaults, though the attribution is not independently tested. Conse-
244 quently, excess-over-control measures *bad recipe versus mediocre recipe*, not versus production. It
245 is an indicative anchor, not a bound, and we decline to publish a corrected point estimate: the cor-
246 rection moves by 5.70pp depending on the anchor set, and we have only 6 rows from two models.

247 8 Limitations

248 **The guarantee is conditional, and absent where it matters most.** Split conformal gives exact
249 coverage under exchangeability between calibration and test. Our calibration set is the set of config-
250 urations a publisher chose to release—a selection event. Conditional on selection, calibration points
251 are not exchangeable with an arbitrary test point and the guarantee does not transfer Barber et al.
252 [2023], Jin and Ren [2024]. It holds for a new checkpoint drawn from the same publication process.
253 It carries **no guarantee** for a recipe the reader tuned themselves and that no publisher would have
254 released—which is the case a practitioner most wants covered.

255 **Accuracy is not the whole of deployment risk.** Tong et al. [2026] show compression frequently
256 decouples accuracy from uncertainty: a model can hold accuracy while its prediction sets inflate. A
257 scheme this work calls low-risk on accuracy may still degrade uncertainty quality. We measure none
258 of that.

259 **One publisher, one toolchain, and why.** See §3. Our own control arm (§7) is direct evidence
260 that practitioner and calibration volume can dominate scheme choice. We measured how confining
261 this is: surveying 10 publishers and 260 quantized model cards, 10 of 222 cards from the other 9
262 publishers carry paired before/after evaluations, 9 of them Intel’s, so others do publish them. What

the other publishers almost never provide is a *Recovery* column, which is what lets each row be checked against 100 $\times$ after/before and rejected on disagreement: of their rows, 6 pass that gate, all from 1 card, against 147 from the control publisher. The corpus is single-publisher not because others publish nothing, but because the control publisher’s cards are, by a wide margin, the ones published in a self-verifying form.

What can and cannot be re-run. The supplement recomputes every tagged figure from the data files it ships. It cannot re-run the faulted-configuration experiments (they need a GPU; the recorded results are included), the harvest from raw model cards (the cards are not redistributed), or the prospective-card checks (those files are not included), and re-running the low-rank comparison mentioned in the supplement needs a checkout of BenchPress’s released code.

An attempt at re-running the calibration-data test. We attempted to reproduce Red Hat’s published W4A16 checkpoints for two of the corpus’s most-measured base models, Qwen2.5-0.5B and Qwen2.5-1.5B, using the `llm-compressor` recipe at its documented defaults, in order to test whether the roughly $171\times$ calibration-data ratio between our own control and production defaults accounts for the observed excess damage. Neither of these two cards pins a library version or names a calibration dataset; both record only the resulting weight representation (symmetric per-group INT4, group size 128 on 1.5B and group size 64 on 0.5B). Neither attempt succeeded. A first run at the current `llm-compressor + compressed-tensors + transformers` stack produced quantized models scoring at or below four-way random chance on 5-shot MMLU (23–28% absolute), well past the worst outcome anywhere in the corpus or the control arm, which we treat as a stack-level anomaly rather than a calibration-data finding. A second run at an era-appropriate pin inferred from the card wording (`llmcompressor==0.4.1, transformers==4.44.2`) could not resolve to a working import: the pinned `llmcompressor` requires a class that was only added to `transformers` in a later release, so the pin does not exist as a working combination. The calibration-data hypothesis therefore remains what the Results section already calls it, consistent with the observation but not independently tested. That the recipe recorded on these two cards is not sufficient to reproduce even the working-stack question, let alone the answer, is directly consistent with the sparsity of documentation reported in §3.

Rows are not independent. Up to sixteen benchmark rows come from a single quantization run. We mitigate with group-conditional calibration and the support floor, and note that a bootstrap over the 19 quantized checkpoints gives [83.2, 94.7]%, close to the row-level interval, but the unit of analysis remains a row. The leave-family-out coverage measurement has the same dependence: each row is scored once under each of 7 calibration families, so the 5719 evaluations are not independent, and every coverage judgment in this paper resamples whole checkpoints rather than counting evaluations.

9 Audit

We record errors found in our own work, including those that changed published claims. This section exists because the alternative—reporting only what survived—would misrepresent how the result was obtained.

Errors that changed a published number.

- *A retracted headline.* An early claim that held-out models came from families never used in construction was wrong; they were new checkpoints of families already present. The strict figure is 118/131.
- *Two parser bugs.* A benchmark label typo cost 11 training rows; separately, MATH-500 was matched as `Math-Lvl-5`, contaminating 7 test rows with scores from a different benchmark.
- *A centring mismatch.* Calibration measured residuals about the stratum mean while intervals centred on the scheme mean, mis-sizing every calibrated width. Found only by independent reimplementations.

- 312 • *A column-order fault.* One card template orders columns [recovery, base, quant], which our
313 parser turned into a spurious -68pp delta. The integrity gate rejected 10 of 11 rows on that
314 card, passing only the one where base = quant made the check degenerate.
- 315 • *A retracted single-publisher claim.* We had asserted that one publisher was close to the
316 only source of paired evaluations. A census of 10 publishers found 10 paired-evaluation
317 cards outside the control publisher, 9 of them Intel’s; what differs is whose rows can be
318 verified (§8).
- 319 • *A bootstrap on the wrong statistic.* The checkpoint-level bootstrap in our coverage audit re-
320 sampled the two-sided indicator while the verdict was judged on the one-sided statistic. An
321 external commenter asked for it to wrap the one-sided figure, which exposed the mismatch
322 (§4).
- 323 • *Inflated pair counts.* An earlier version of the tool reported the number of (test row, cali-
324 bration family) evaluations as if it were the number of rows: with 8 families each row is
325 scored 7 times, so 5719 evaluations came from only 817 rows. Every row-count claim was
326 recomputed on `distinct_rows`.
- 327 • *Two levels judged differently.* Cells were judged on the one-sided coverage statistic and
328 refused when the point estimate fell below the threshold, but schemes had been judged on
329 the two-sided statistic and refused only when the entire bootstrap interval fell below it. A
330 single classifier now judges both levels the same way, and the envelope’s refused list is a
331 computed copy of the ranking tool’s, checked by a test.

332 **A regression we nearly shipped.** An audit indicated that a raw asymmetric empirical band cov-
333 ered better than the shipped symmetric conformal interval (93.4% versus 89.6%). Since quantization
334 damage is left-skewed, a symmetric band is obviously the wrong shape, and we were about to switch.
335 On measuring first, the empirical band returns an *infinite* interval on 43% of rows, because a two-
336 sided empirical index requires more calibration points than are often available. An infinite interval
337 covers by construction. On rows where both are finite, the original wins on coverage *and* width. The
338 change was reversed.

339 **An overclaimed guarantee.** The artifact was described as carrying a coverage guarantee without
340 qualification. Checking the method against the conformal literature rather than against our own
341 earlier work showed the guarantee is conditional on exchangeability that selection breaks. Corrected
342 in §8.

343 **Provenance discipline.** Two parser fixes were informed by inspecting rows that were already in
344 the test set. Rather than silently retain them, we record which fixes were test-informed and exclude
345 the affected family from the strict headline.

346 **An attempted extension of the adversarial arm to 3B.** On a fresh 40 GB A100 with a fresh
347 dependency install, we re-ran the adversarial sweep to (i) verify that yesterday’s numbers repro-
348 duce and (ii) add a Qwen2.5-3B-Instruct row. The regression check passed: base evaluations
349 on Qwen2.5-0.5B-Instruct and SmolLM-135M-Instruct reproduced byte-for-byte, deterministic-
350 quantization recipes (per-tensor RTN, w8a8 without smoothing) reproduced byte-for-byte, and
351 stochastic GPTQ recipes stayed within $\leq 0.85\text{pp}$ of yesterday, consistent with fp16 Hessian-
352 accumulation noise. The 3B extension did not land: base evaluations on Qwen2.5-3B-Instruct
353 completed, but all five quantization recipes hit `torch.cuda.OutOfMemoryError` during the paired-
354 evaluation forward pass at batch size 2, not during quantization itself; the fp16 model weights
355 (≈ 6 GB), PyTorch’s caching allocator fragmentation from unloaded prior states, and activation
356 memory for a batch of 2 long sequences together exceeded the 40 GB budget with roughly 10 GB
357 of free memory needed for the next allocation. A single 80 GB card would be expected to accom-
358 modate this, so the constraint is hardware, not methodology. The adversarial arm therefore remains
359 bounded at $\leq 1.5\text{B}$ as before; §8 states this scope directly.

360 **Mechanical verification.** The figures this paper reports as results are macros generated from a
361 claims registry that recomputes each value from source artifacts, and the accompanying code fails
362 a test if an outward-facing document contains an untraced numeric literal. The exceptions are his-
363 torical figures in the error list above (row counts and the -68pp delta), which are quoted from the
364 provenance record rather than recomputed.

10 Conclusion

Per-model prediction of quantization accuracy damage from card-level features does not work on published data, and the reason is largely measurement noise rather than model capacity; predictors that use checkpoint-level statistics were not tested and remain open. What does work is narrower: a per-scheme calibrated envelope that is validated at 90.1% two-sided coverage on unseen checkpoints, of families already present in training, for 4 of the 6 schemes, and that says which of its own cells it cannot judge. Each cell is trusted; or insufficient evidence, in which case the interval is still shown but flagged and the scheme demoted; or refused, in which case the interval is withheld. Today 9 of 17 cells are insufficient evidence and 0 are refused. The more useful contribution may be the limitation itself—the published record of quantized models is outcome-truncated, and we measured, on deliberately faulted recipes for models up to 1.5B, losses more than four times worse than anything it contains. A risk estimate built on that record is a floor on risk, not a ceiling, and should say so.

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